Cultural Bias Matters: MultiMM, a Cross-Cultural Benchmark, and SEMD, a Sentiment-Enriched Model for Multimodal Metaphor Understanding

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Abstract

Multimodal metaphor understanding remains heavily shaped by English-centric resources, even though metaphor interpretation is known to vary with culture and with the interaction between text, image, and affect. We present MultiMM, a cross-cultural benchmark of 8,461 advertisement text-image pairs spanning Chinese and English, annotated for metaphor occurrence, source and target domains, and three-way sentiment polarity. MultiMM supports two benchmark tasks, metaphor detection and sentiment analysis, and is accompanied by a broad evaluation suite covering textual, visual, multimodal, and multimodal large language model baselines. We also introduce SEMD, a simple sentiment-enriched multimodal model that combines multilingual BERT text features, ViT image features, and auxiliary sentiment representations for metaphor detection. Across 22 baselines, SEMD achieves the best macro-F1 on both languages for metaphor detection (80.16 on English and 77.79 on Chinese) and sentiment analysis (75.69 on English and 70.51 on Chinese). Ablation and translation-based transfer experiments show that sentiment features consistently strengthen metaphor recognition and that performance drops under cross-cultural transfer remain substantial. The benchmark and analyses expose cultural bias in current multimodal metaphor systems and provide a concrete testbed for developing more robust cross-cultural models.

1 Introduction

Metaphor is a central mechanism for expressing abstract ideas through more concrete concepts, and it is deeply intertwined with perception, persuasion, and culture (Lakoff, 1994; Kövecses, 2010; Ibarretxe-Antuñano, 2013). In advertising, metaphor is often explicitly multimodal: textual slogans, visual symbols, and affective framing jointly communicate an intended message (Forceville, 2002; Forceville and Urios-Aparisi, 2009; Forceville, 2017). This makes multimodal metaphor understanding an important problem for computational semantics, multimedia analysis, and cross-cultural NLP.

Recent metaphor processing research has made steady progress on textual detection and interpretation (Shutova et al., 2010; Liu et al., 2020; Ge et al., 2022), and newer datasets have pushed metaphor research into multimodal settings (Zhang et al., 2021; Xu et al., 2022; Akula et al., 2023; Lu et al., 2025). However, the available resources remain fragmented across languages, domains, and tasks. Existing multimodal datasets are predominantly English-only, Chinese-only, or focused on memes and visual metaphor rather than cross-cultural advertisements. As a result, current models can appear robust while still relying on culture-specific regularities.

This gap matters because metaphor conventions are not culturally uniform. Conceptual mappings, source-domain preferences, and rhetorical framing differ across social and linguistic communities (Kövecses, 2010; Ibarretxe-Antuñano, 2013). Sentiment adds another important dimension: metaphors are often emotionally salient (Citron and Goldberg, 2014; Mohammad et al., 2016), and affective cues may provide a partial bridge even when source-domain realizations differ across cultures. These observations motivate a benchmark that explicitly tests cross-cultural robustness and a model that incorporates sentiment as a supporting signal.

In this paper, we introduce **MultiMM**, a cross-cultural benchmark of 8,461 Chinese and English advertisements, and **SEMD**, a sentiment-enriched multimodal baseline tailored to this benchmark. MultiMM contains nearly balanced Chinese and English subsets and provides labels for metaphor occurrence, target domain, source domain, and sentiment polarity. It supports two benchmark tasks: metaphor detection and sentiment analysis. SEMD

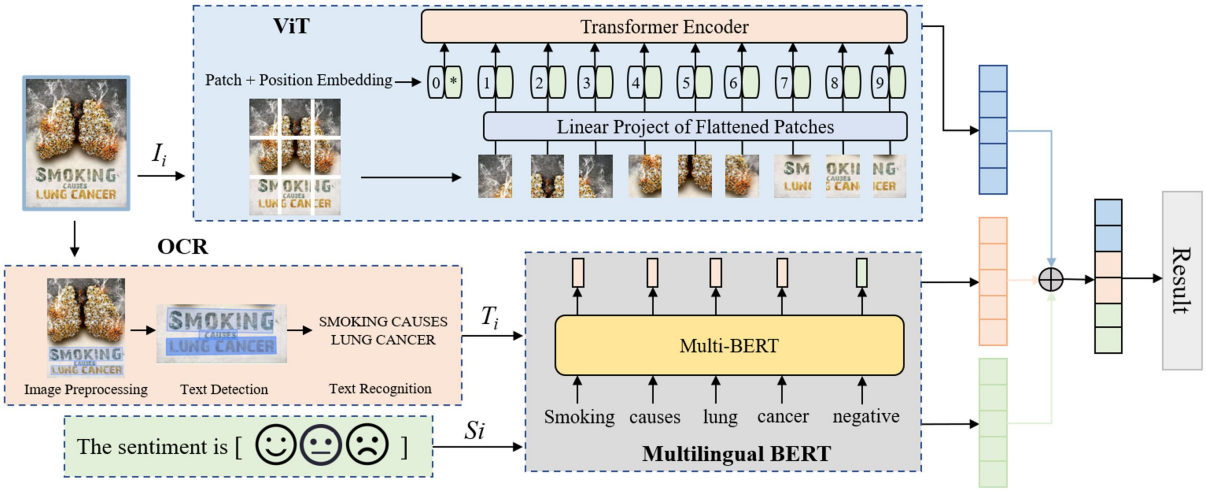

Figure 1: Overview of SEMD. The model uses three branches: ViT image features, multilingual BERT text features from OCR-extracted advertisement text, and an auxiliary sentiment representation. The three vectors are fused by concatenation for metaphor detection; for sentiment analysis, SEMD uses the image and text branches only.

combines frozen multilingual text and vision encoders with auxiliary sentiment features, giving a strong but simple baseline for measuring the value of affective information in multimodal metaphor recognition.

Our main contributions are as follows:

- We introduce MultiMM, to our knowledge the first large-scale cross-cultural benchmark for multimodal metaphor understanding in advertisements, with 8,461 Chinese-English text-image pairs and fine-grained annotations for metaphor, source domain, target domain, and sentiment.
- We establish a rigorous benchmark protocol with two tasks and 22 baselines spanning textual, visual, multimodal, and multimodal large language model systems, plus translation-based transfer evaluation to probe cultural robustness.
- We propose SEMD, a sentiment-enriched multimodal model that consistently improves over strong baselines on both languages and both tasks, showing that affective features are useful auxiliary signals for multimodal metaphor understanding.

2 Related Work

Computational metaphor processing. Early computational work on metaphor focused primarily on textual detection and interpretation (Shutova et al., 2010, 2013; Liu et al., 2020). More recent models incorporate contextualized encoders and explicit conceptual structure (Choi et al., 2021; Ge et al., 2022). These studies show that pretrained language models help metaphor identification, but they are still largely grounded in textual input and predominantly English benchmarks.

Multimodal metaphor resources and models. Multimodal metaphor has long been studied in linguistics and advertising analysis (Forceville, 2002; Forceville and Urios-Aparisi, 2009; Forceville, 2017). Computational datasets such as MultiMET (Zhang et al., 2021), MET-Meme (Xu et al., 2022), MetaCLUE (Akula et al., 2023), and EmoMeta (Lu et al., 2025) expanded the scope of multimodal metaphor research across advertisements, memes, and visual reasoning. On the modeling side, multimodal metaphor detection has used visual-text fusion, concreteness-aware modeling, and more elaborate multimodal interactions (Su et al., 2021; He et al., 2024b,a). Yet these resources usually emphasize a single language or a single culture, making it difficult to distinguish robust metaphor understanding from dataset-specific pattern matching.

Culture, affect, and cross-cultural evaluation. Metaphorical framing is culturally situated (Kövecses, 2010; Ibarretxe-Antuñano, 2013), and advertisements are an especially culture-sensitive setting because source-domain choices often draw on local symbols and persuasive conventions. At the same time, metaphors are closely tied to emotion and affective intensity (Citron and Goldberg, 2014;

Mohammad et al., 2016). Recent benchmarks for LLM metaphor understanding (Tong et al., 2024) and large-scale multimodal models such as GPT-4o, Gemini, LLaVA, and Qwen2.5-VL (Hurst et al., 2024; Team et al., 2023; Liu et al., 2023; Bai et al., 2025) make it natural to ask whether broad multimodal capabilities transfer to culturally grounded metaphor tasks. MultiMM is designed to answer that question directly.

3 MultiMM: A Cross-Cultural Benchmark

3.1 Data collection and filtering

MultiMM contains 8,461 advertisement text-image pairs: 4,397 Chinese examples and 4,064 English examples. Chinese advertisements were collected through keyword-based Baidu retrieval over products, public-service campaigns, and common metaphor domains, supplemented with an iFlytek advertisement set. English examples were collected from a public advertisement resource covering commercial products and public-service announcements. We removed duplicates using MD5 hashing (Rivest, 1992), filtered out non-advertisements, blurred instances, and images smaller than 350×350, and extracted textual content with OCR followed by manual correction. For OCR we used a practical PP-OCR pipeline (Du et al., 2020) as an initial text extractor.

3.2 Annotation schema and protocol

Each instance is annotated with an *AnnotationModel* tuple $(Occurrence, Target, Source, Sentiment)$, where *Occurrence* indicates whether the advertisement is metaphorical, *Target* and *Source* denote the target and source domains, and *Sentiment* is a three-way polarity label in {negative, neutral, positive}. Visual domains were verbalized during annotation so that domain mappings could be described in a shared semantic form across modalities.

The guidelines were derived from multimodal metaphor research in advertising (Forceville, 2002; Forceville and Urios-Aparisi, 2009). Chinese data were annotated by five expert annotators and English data by three expert annotators. The process used small-group annotation with adjudication, together with training rounds and iterative guideline refinement. To reduce monocultural bias, the annotation process incorporated multicultural feedback during guideline revision. Inter-annotator agreement measured by Fleiss' κ (Fleiss, 1971) was 0.73 for metaphor occurrence, 0.70 for target domain, 0.66 for source domain, and 0.82 for sentiment, indicating substantial agreement overall.

Figure 2: Annotated example from MultiMM. Each advertisement is paired with OCR-corrected text and labels for metaphor occurrence, target domain, source domain, and sentiment polarity.

3.3 Benchmark tasks and statistics

MultiMM supports two benchmark tasks. The first is **metaphor detection**, a binary classification problem over metaphorical versus literal advertisements. The second is **sentiment analysis**, a three-way classification problem over negative, neutral, and positive sentiment. We release fixed train, validation, and test splits with 6,768, 846, and 847 instances, respectively.

Table 1 summarizes the benchmark. The dataset is close to balanced across languages and contains a substantial number of metaphorical examples in both subsets. A notable difference is text length: Chinese advertisements contain 145,312 words in total with an average of 33 words per sample, whereas English advertisements contain 68,189 words with an average of 15 words. This difference makes MultiMM useful not only for culture-sensitive evaluation but also for testing robustness to denser textual context in Chinese advertisements.

Design rationale. We made three benchmark design choices intentionally. First, we focus on advertisements because they offer naturally occurring multimodal metaphors in which text, imagery, and persuasion are tightly coupled. Second, we keep the Chinese and English subsets close in size so that cross-cultural comparisons are not dominated by sample imbalance. Third, we annotate source and target domains in addition to benchmark labels. Although the present paper evaluates only

Item	CN	EN	Total
Total	4,397	4,064	8,461
Metaphorical	2,583	2,189	4,772
Literal	1,814	1,875	3,689
Total Words	145,312	68,189	213,501
Average Words	33	15	24
Training Set Size	3,517	3,251	6,768
Validation Set Size	440	406	846
Test Set Size	440	407	847

Table 1: Statistical overview of MultiMM. CN refers to Chinese data, while EN refers to English data.

metaphor detection and sentiment analysis, the domain annotations make MultiMM useful for future work on metaphor interpretation, explanation, and culturally grounded concept mapping.

4 SEMD: Sentiment-Enriched Multimodal Detection

SEMD is a simple multimodal fusion model designed to test whether affective features improve cross-cultural metaphor recognition. The model follows the architecture in Figure 1 and uses frozen pretrained encoders for the two primary modalities.

Text and image encoders. For text, SEMD uses multilingual BERT (`bert-base-multilingual-cased`) (Devlin et al., 2019). For images, it uses ViT-B/16 (`vit-base-patch16-224`) (Dosovitskiy et al., 2020). Both encoders produce 768-dimensional representations. The textual input is the OCR-extracted advertisement text, truncated to 30 tokens.

Sentiment representation. For metaphor detection, SEMD adds a third 768-dimensional sentiment vector produced by an auxiliary sentiment-analysis component. The design intuition is that sentiment provides an additional semantic cue for interpreting figurative advertisements, especially when source-domain realizations differ across cultures but affective framing is still informative.

Fusion and prediction. For metaphor detection, the three modality vectors are concatenated:

$$\mathbf{h}_m = [\mathbf{h}_{text}; \mathbf{h}_{img}; \mathbf{h}_{sent}],$$

and passed to a feed-forward classifier with dropout. For sentiment analysis, SEMD uses image and text features only:

$$\mathbf{h}_s = [\mathbf{h}_{text}; \mathbf{h}_{img}].$$

In both cases, the fused representation is mapped to task labels with a cross-entropy objective.

Training details. SEMD is trained with AdamW (Loshchilov and Hutter, 2018) in PyTorch (Paszke et al., 2019). We keep the pretrained encoders frozen and optimize the fusion head for 10 epochs with batch size 64, dropout 0.3, and learning rates searched in the range $[3 \times 10^{-5}, 5 \times 10^{-4}]$.

5 Experimental Setup

Metrics. The primary metric is macro-F1, which is appropriate for both the binary metaphor task and the three-way sentiment task. We also report accuracy and macro-precision in the translation-based transfer experiment.

Baselines. We compare against 22 baselines grouped into four families: textual models, visual models, multimodal models, and multimodal large language models. The textual group includes mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), SixT+ (Chen et al., 2022), LaBSE, and several large language models. The visual group includes VGG16 (Simonyan and Zisserman, 2014), ResNet50 (He et al., 2016), and ViT (Dosovitskiy et al., 2020). The multimodal group includes conventional fusion baselines and MLLMs such as GPT-4o (Hurst et al., 2024), LLaVA (Liu et al., 2023), Gemini (Team et al., 2023), and Qwen2.5-VL (Bai et al., 2025). This setup allows us to test whether recent general-purpose MLLMs can match task-specific multimodal models on culturally grounded metaphor understanding.

The baseline suite is deliberately heterogeneous. It includes unimodal models that reveal how much signal is available from text or image alone, conventional multimodal models that test whether explicit fusion is sufficient, and instruction-tuned MLLMs that probe whether broad world knowledge transfers to figurative advertisement under-

standing. This makes the benchmark useful not only for leaderboard comparison but also for diagnosing what kinds of multimodal inductive bias are actually needed.

Because MultiMM labels metaphor and sentiment separately, the same baseline pool can also reveal whether an apparent gain comes from better figurative reasoning or from easier polarity prediction. A method that helps only sentiment but not metaphor is therefore still informative: it shows that the benchmark is discriminating between related but non-identical capabilities.

Cross-cultural transfer. Beyond standard in-language evaluation, we also test translation-based robustness by translating English advertisements into Chinese and Chinese advertisements into English, then rerunning metaphor detection. This probes whether a model preserves metaphor judgments when the linguistic surface form is shifted across cultures.

6 Main Results

Table 2 reports the main results. SEMD achieves the best overall macro-F1 on all four settings: 80.16 on English metaphor detection, 77.79 on Chinese metaphor detection, 75.69 on English sentiment analysis, and 70.51 on Chinese sentiment analysis.

Several findings stand out. First, multimodal modeling is clearly necessary. The best textual models remain well below the best multimodal systems on metaphor detection: XLM-R reaches 69.90 on English and LaBSE reaches 66.41 on Chinese, while SEMD reaches 80.16 and 77.79. The best visual-only baseline, ViT, is stronger than most text-only models on metaphor detection, indicating that the visual rhetoric of advertisements carries substantial metaphor signal. Still, vision alone is insufficient, and the best performance requires joint reasoning over text and image.

The contrast between tasks is also informative. Image-only baselines are relatively competitive for metaphor detection, but they remain clearly weaker for sentiment analysis, which suggests that visual substitution often exposes metaphor status while sentiment depends more on the joint reading of slogan and image. MultiMM therefore separates figurative reasoning from coarse affect prediction instead of collapsing them into a single signal.

Second, SEMD improves over strong multimodal baselines on both languages. On metaphor detection, it exceeds the strongest non-SEMD baseline by 1.12 F1 points on English (80.16 versus 79.04 for CMGCN) and 2.26 points on Chinese (77.79 versus 75.53 for CES). On sentiment analysis, it improves over CES by 2.45 points on English and 0.25 points on Chinese. These gains are modest in absolute size but consistent across tasks and languages.

Third, current MLLMs are not competitive on this benchmark. LLaVA is the strongest of the tested MLLMs on English metaphor detection at 73.31 F1, but it still trails SEMD by nearly 7 points. GPT-4o, Gemini, and Qwen2.5-VL perform substantially worse, especially on sentiment analysis and on Chinese data. This suggests that broad multimodal instruction tuning does not automatically yield reliable cross-cultural metaphor understanding.

7 Analysis

7.1 Ablation of sentiment and fusion

Table 3 isolates the contribution of the sentiment branch and the fusion strategy for metaphor detection. Concatenation is the strongest fusion method both with and without sentiment. More importantly, adding sentiment features consistently helps every fusion strategy. Under concatenation, sentiment improves F1 from 78.64 to 80.88 on English and from 74.84 to 77.39 on Chinese, gains of 2.24 and 2.55 points, respectively. This supports the core hypothesis behind SEMD: affective information is not merely correlated with metaphor usage, but useful as an auxiliary signal for recognizing multimodal metaphor.

7.2 Cross-cultural transfer via translation

Table 4 shows the translation-based transfer experiment. SEMD remains the strongest system after bidirectional Chinese-English translation, reaching 77.64 F1 for English-to-Chinese transfer and 75.53 F1 for Chinese-to-English transfer. The absolute scores are lower than standard in-language evaluation, which indicates that metaphor recognition does not transfer perfectly under translation even when the advertisement content is preserved. At the same time, the gap between SEMD and the two comparison models shows that multimodal grounding and sentiment enrichment improve robustness under linguistic transfer.

Model	Metaphor Detection F1 (%)		Sentiment Analysis F1 (%)	
	EN	CN	EN	CN
Random	46.70	41.20	40.41	35.02
mBERT	64.00	65.52	62.83	58.43
XLM_R	69.90	62.38	69.44	66.21
SixTP	68.61	65.60	64.43	65.59
LaBSE	68.15	66.41	69.93	69.02
GPT-3.5-TURBO	30.95	15.49	42.06	52.08
Llama2-8B	22.58	19.73	40.11	50.72
Llama3.1-70B	44.04	49.70	33.46	30.90
Deepseek R1-70B	44.06	47.95	53.38	49.01
VGG16	70.64	67.85	59.90	59.02
ResNet50	71.46	68.56	57.49	58.86
ViT	71.67	69.04	61.46	62.17
mBERT-Res	77.83	74.40	69.75	68.84
CMGCN	79.04	74.91	72.79	70.19
CES	78.45	75.53	73.24	70.26
MET	76.04	72.51	71.95	67.17
MFC	75.84	73.25	71.93	69.60
GPT-4O	64.00	67.00	36.00	29.00
LLaVA	73.31	69.84	56.72	53.21
Gemini	64.19	68.51	53.61	43.48
Qwen2.5-VL-3b	57.30	41.69	52.50	55.26
Qwen2.5-VL-7b	56.62	54.51	50.71	55.15
Qwen2.5-VL-72B	59.12	67.66	50.44	57.17
SEMD	**80.16**	**77.79**	**75.69**	**70.51**

Table 2: Experimental results on metaphor detection and sentiment analysis tasks. The best overall results are highlighted in **bold**, while the best results within each modality are underlined. Models are categorized by modality, with textual, visual, and multimodal models highlighted in cyan, green, and orange, respectively.

Sentiment Feature	Fusion Method	EN F1 (%)	CN F1 (%)
w/o	add	75.95	72.05
	max	76.29	74.09
	concat	78.64	74.84
w/	add	77.76	74.37
	max	78.59	74.37
	concat	**80.88**	**77.39**

Table 3: Results of the ablation study. The best and second-best results are highlighted in **bold** and underline, respectively.

7.3 What the benchmark reveals

The analysis figures help explain why MultiMM is a useful diagnostic benchmark. Figure 3 shows clear cross-cultural differences in sentiment usage. Among metaphorical advertisements, positive sentiment dominates English data (74.75%) more strongly than Chinese data (56.42%), while Chinese metaphorical advertisements have a much larger neutral share (40.41% versus 24.57%). Literal advertisements are also more neutral in Chinese (52.91%) than in English (39.28%). These differences help explain why sentiment is informative but not sufficient: it provides a stable auxiliary cue, yet its distribution is still culture-dependent.

Figure 4 highlights a second challenge: source-domain vocabularies cluster differently across languages. The English subset includes concentrated regions around sources such as *lion* and *car*, whereas the Chinese subset features culturally grounded sources such as *loong*, *panda*, and other nature-oriented symbols. Together with the longer average Chinese text length in Table 1, this helps explain why Chinese performance remains lower than English performance across most systems.

Figures 5 and 6 refine this picture. The Chinese subset contains recurring symbolic sources tied to local convention, and the word clouds show that source choice also shifts with sentiment. Shared high-frequency sources remain visible across settings, but their prominence changes by language and polarity, which makes MultiMM useful for diagnosing when a model has learned broad metaphor cues versus culture-specific shortcuts.

Overall, the benchmark reveals two complemen-

		mBERT	mBERT-Res	SEMD
EN → CN	Acc (%)	64.29	76.35	78.57
	Pre (%)	63.72	76.73	78.19
	F1 (%)	59.70	75.63	77.64
CN → EN	Acc (%)	65.22	73.86	75.45
	Pre (%)	67.46	75.90	77.73
	F1 (%)	65.31	73.96	75.53

Table 4: The results of metaphor detection tasks after Chinese-English translation. Experimental results on metaphor detection task via bidirectional Chinese-English translation.

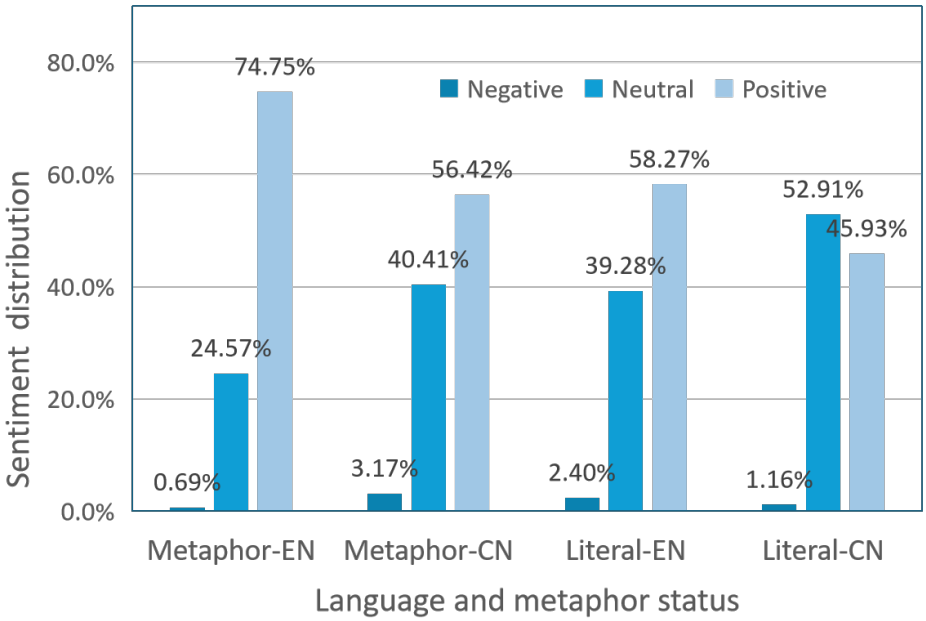

Figure 3: Sentiment distributions for metaphorical and literal advertisements in Chinese and English. English metaphorical advertisements are more strongly skewed toward positive sentiment than Chinese metaphorical advertisements.

tary facts. Current multimodal systems do benefit from strong visual encoders and sentiment-aware fusion, but cross-cultural metaphor understanding remains unsolved. Performance gains on English do not guarantee equivalent robustness on Chinese, and translation does not eliminate the gap. MultiMM therefore serves not only as an evaluation resource but also as a diagnostic framework for studying cultural bias in multimodal semantics.

8 Conclusion

We presented MultiMM, a cross-cultural benchmark for multimodal metaphor understanding in Chinese and English advertisements, and SEMD, a sentiment-enriched multimodal baseline for the benchmark. MultiMM contributes broad coverage, fine-grained annotations, and an evaluation protocol that exposes meaningful cultural differences in metaphor and sentiment distributions. SEMD shows that a simple integration of multilingual text, vision, and sentiment signals yields strong and consistent improvements over textual, visual, multimodal, and MLLM baselines. The results also make clear that cross-cultural metaphor understanding is still difficult, particularly under translation and on Chinese data. We hope MultiMM will support more rigorous work on culturally robust multimodal semantics.

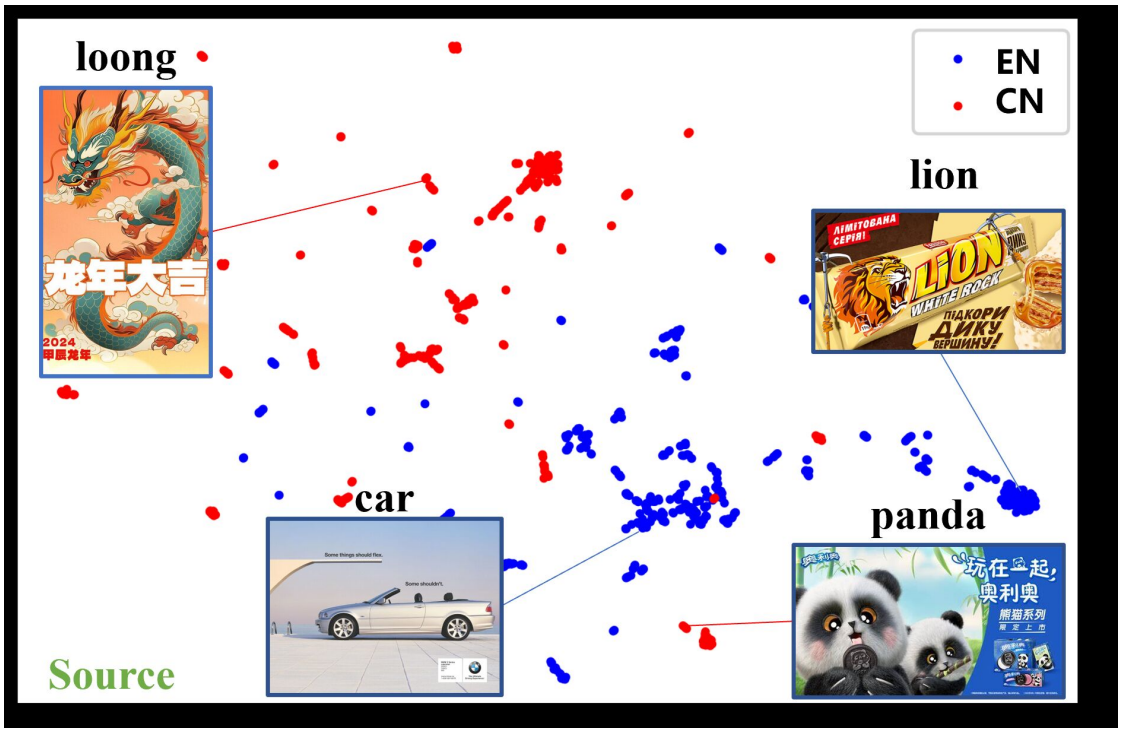

Figure 4: Cross-cultural distribution of source-domain vocabulary. English and Chinese advertisements form visibly different source-domain clusters, with culturally specific symbols appearing prominently in the Chinese subset.

Figure 5: Frequent Chinese source motifs in MultiMM, including *fu*, *loong*, cloud, and magpie imagery.

Figure 6: Source-domain word clouds by language and sentiment. Shared sources such as *water*, *ice*, and *tree* coexist with culture-specific terms that are more prominent in Chinese.

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